

TEXTGAMES: Learning to Self-Play Text-Based Puzzle Games via Language Model Reasoning

Anonymous ACL submission

Abstract

Reasoning is a fundamental capability of large language models (LLMs), enabling them to comprehend, analyze, and solve complex problems. In this paper, we introduce TEXTGAMES, an innovative benchmark specifically crafted to assess LLMs through demanding text-based games that require advanced skills in pattern recognition, spatial awareness, arithmetic, and logical reasoning. Our analysis probes LLMs’ performance in both single-turn and multi-turn reasoning, and their abilities in leveraging feedback to correct subsequent answers through self-reflection. Our findings reveal that, although LLMs exhibit proficiency in addressing most easy and medium-level problems, they face significant challenges with more difficult tasks. In contrast, humans are capable of solving all tasks when given sufficient time. Moreover, we observe that LLMs show improved performance in multi-turn predictions through self-reflection, yet they still struggle with sequencing, counting, and following complex rules consistently. Additionally, models optimized for reasoning outperform pre-trained LLMs that prioritize instruction following, highlighting the crucial role of reasoning skills in addressing highly complex problems.

1 Introduction

Reasoning is a fundamental skill essential for logical thinking and development, enabling large language models (LLMs) to tackle complex problems (Wei et al., 2022; Longpre et al., 2023; Srivastava et al., 2023). This skill emphasizes the need for creating LLMs capable of handling tasks such as mathematical (Hendrycks et al., 2021; Shao et al., 2024; Trinh et al., 2024), commonsense (Talmor et al., 2019; Geva et al., 2021; Brohan et al., 2023), and symbolic reasoning (Nye et al., 2021; Sprague et al., 2024). In general, reasoning is a multifaceted ability that involves understanding the context and

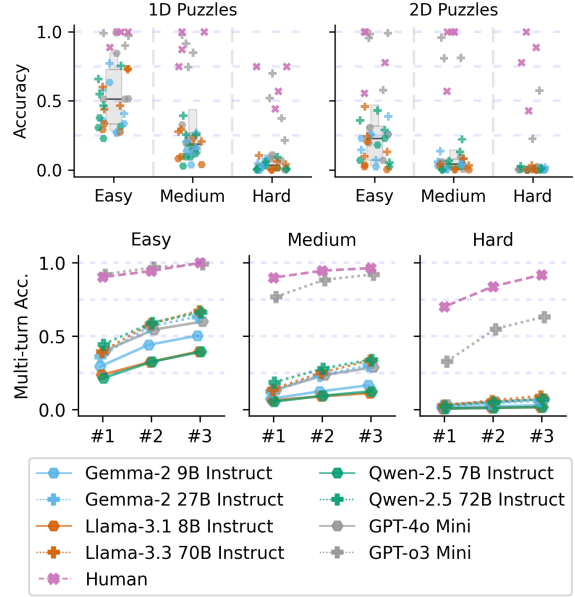


Figure 1: Single-turn performance on TEXTGAMES games across 1D and 2D Puzzles challenges with varying difficulty levels (**top**), alongside the improvement in accuracy achieved through increased turn attempts via self-reflection, with the x-axis representing the number of turns (**bottom**).

effectively applying inference to solve problems. Research on LLMs has examined their reasoning capabilities across various dimensions, including their capacity to follow instructions for multi-hop reasoning (Yang et al., 2024b), comprehend psychological concepts (Almeida et al., 2024), and use context in classification tasks (Winata et al., 2024), and constrained logical tasks (Zhou et al., 2023). LLMs have also demonstrated remarkable skills in game reasoning, such as solving crossword puzzles (Berruti et al., 2024; Saha et al., 2024; Zugarini et al., 2024), physics-based puzzle games (Oh et al., 2024), and turn-based games (Feng et al., 2024; Guo et al., 2024).

A longstanding issue in reasoning with LLMs is their tendency to hallucinate and inconsistency dur-

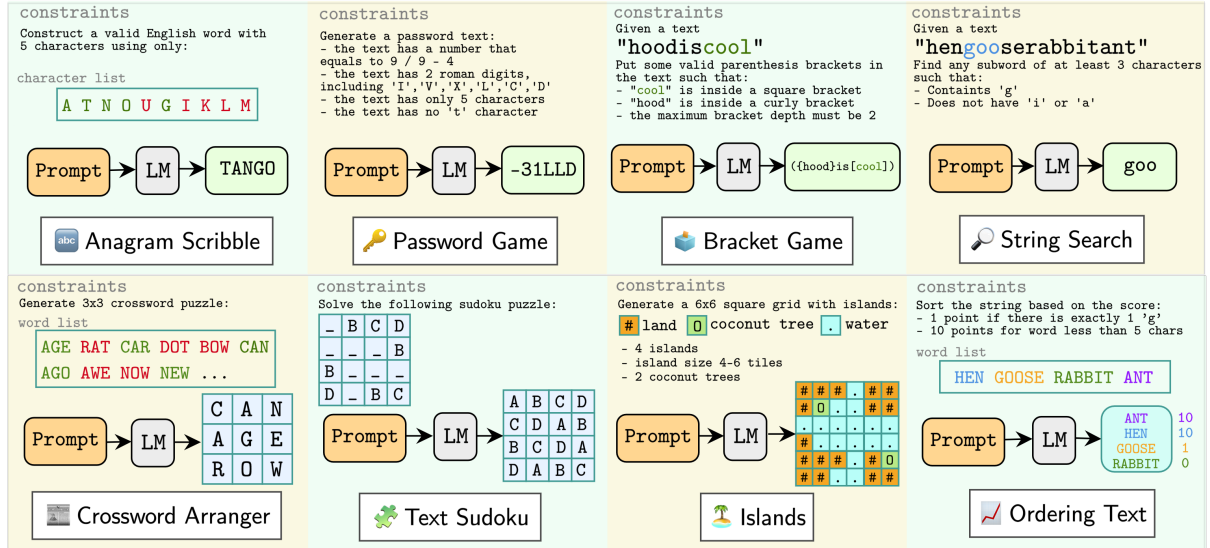


Figure 2: TEXTGAMES BENCHMARK consists of eight text-based puzzle games, each with unique constraints and gameplay mechanics. The top four games are 1D Puzzles, while the bottom four are 2D Puzzles.

ing inference (Maynez et al., 2020; Ji et al., 2023a; Huang et al., 2024b). Recently, self-reflection techniques have been employed to mitigate these hallucinations and improve the performance of LLMs through multiple rounds of follow-up interactions (Ji et al., 2023b). Additionally, self-evaluation has been applied to question-answering tasks (Ren et al., 2023), offering feedback that enables models to correct themselves. Consequently, LLMs have demonstrated the ability to rectify errors across various domains, gradually producing correct answers over successive iterations (Shinn et al., 2024). Despite these advancements, we aim to further challenge LLMs by engaging them with puzzles that require a combination of skills, including pattern recognition, spatial awareness, arithmetic, and logical thinking.

In our work, we introduce TEXTGAMES, a new benchmark designed to assess the proficiency of LLMs in solving text-based logical puzzle games and performing complex, constraint-based reasoning. The intricate rules of these puzzles allow us to evaluate the LLMs’ capacity to follow detailed instructions. Additionally, we investigate whether LLMs can self-reflect on their previous generations when given feedback, correcting their errors by responding to specific error messages and refining their outputs. We also provide performance comparison between reasoning-specialized LLMs, with models that emphasize instruction-following. Our analysis indicates that even the recent advanced LLMs, such as the Llama 70B (Dubey et al., 2024)

and Qwen2 72B Instruct (Yang et al., 2024a) models, perform adequately on Easy and Medium levels but struggle at the Hard level. In contrast, models specifically optimized for reasoning, like GPT-o3 Mini, exhibit strong performance on these more difficult tasks, as illustrated in Figure 1. We hypothesize that this disparity arises because TEXTGAMES demands a high level of reasoning ability to comprehend the rules and apply a combination of reasoning skills to solve the problems that Instruct models may not fully possess.

In summary, our contributions are threefold:

- We introduce TEXTGAMES,¹ a text-based game benchmark that assesses LLMs’ various logical reasoning skills. The benchmark features eight puzzle games across three difficulty levels. Figure 2 offers an overview of the game visualizations.
- We perform a thorough evaluation across a range of LLMs, including both off-the-shelf and proprietary models, in zero-shot and one-shot scenarios. We additionally compare their performance with that of human participants.
- We demonstrate that LLMs improve when given feedback in multi-turn interaction, enabling them to self-reflect on previous generations. Our observation on reasoning-focused models’ performance also reveals that there can be diminishing returns on test-time scaling in some difficult games.

¹The code will be released after acceptance.

2 TEXTGAMES BENCHMARK

We introduce our benchmark TEXTGAMES, which comprises eight text-based puzzle games, each featuring three distinct levels of difficulty, aimed at evaluating the reasoning abilities of LLMs. These games are meticulously designed to assess a wide array of reasoning skills, encompassing both abductive and deductive reasoning. Additionally, we differentiate various skills through diverse output formats, as described in Table 1.

2.1 List of Games

We provide a detailed definition of the games as follows:

2.1.1 Anagram Scribble

Given a list of Latin characters, the player’s objective is to arrange them into a valid N -character English word, without regard to case sensitivity. We explore two scenarios: one where characters can be used multiple times and another where each character can only be used once.

2.1.2 Password Game

Given a set of rules, the player is challenged to construct a sequence of characters that fulfills all specified requirements, similar to creating a password. These rules involve generating text based on character counts, incorporating English alphanumeric characters, distinguishing between uppercase and lowercase letters, and including special characters and Roman numerals. Additionally, we introduce more complex tasks that require common-sense knowledge, such as identifying the capital city or continent of a specified country. Furthermore, we add simple arithmetic constraints, such as “The text must include a number equal to seven times six.”

2.1.3 Bracket Game

Given a concatenation of several English words, the player is tasked with enclosing segments of the text using four different types of parentheses: ‘[]’, ‘{}’, ‘()’, and ‘<>’. These brackets must be correctly paired where each open bracket must have a corresponding close bracket, and vice versa. Additionally, there are requirements regarding bracket depth that the player must adhere to.

2.1.4 String Search

Given a random sequence of characters mixed with some valid English words, the player is challenged

to find a substring—a consecutive sequence of characters—that meets a specified set of rules. These rules dictate conditions such as the length of the substring, required characters, prohibited characters, and whether the resulting substring must be a palindrome.

2.1.5 Crossword Arranger

Given a list of English words, each of length N , the player is tasked with arranging these words into a crossword puzzle. Without any repetitions, a total of $2N$ words from the list must be placed in either a horizontal or vertical orientation, forming a connected configuration within an $N \times N$ square grid. Blank cells are not used to separate the words.

2.1.6 Text Sudoku

Given a sparsely filled square grid of size $N^2 \times N^2$, the player is tasked with filling the blank cells with numbers such that no identical numbers appear within the same row, column, or $N \times N$ sub-grid. The player must fill only the blank cells, leaving the pre-filled cells unchanged. We utilize grids with N equal to 2 and 3, meaning the numbers range from 1 to 4 and 1 to 9, respectively. Alternatively, these numbers can be substituted with unique characters; for instance, we experiment with using Latin alphabets ‘A’ to ‘I’ in place of numbers 1 to 9.

2.1.7 Islands

Given a grid size of N , along with a specified set of rules, the player must construct an $N \times N$ square grid using the characters ‘.’, ‘#’, or ‘o’, which represent water, land, and coconut trees, respectively. A contiguous group of land tiles connected in the four cardinal directions forms an island. The task requires adherence to all rules, which govern the number of islands, the size of each island, and the allowable number of coconut trees.

2.1.8 Ordering Text

Given a set of scoring rules and a list of words, the player is tasked with sorting the list from the highest-scoring word to the lowest. The scoring rules encompass checks for the presence of specific character sequence patterns, the length of the words, as well as the prefixes and suffixes of the words. Points in each scoring rule can range from -100 to 100 .

2.2 Challenges and Difficulty Levels

For comprehensive details about the games, including formats, categories, and the reasoning skills

Task	Output Format	Category	Skill-sets	Reasoning
1D Puzzles				
Anagram Scribble	Single-line text	English words	Pattern Recognition, Knowledge	Abductive
Password Game	Single-line text	Numbers & Characters	Arithmetic, Knowledge	Abductive
Bracket Game	Single-line text	Coordinates	Counting	Deductive
String Search	Single-line text	String Matching	Logical Thinking	Deductive
2D Puzzles				
Crossword Arranger	2D-Grid	English words	Pattern Recognition, Spatial Awareness	Deductive
Text Sudoku	2D-Grid	Numbers & Characters	Spatial Awareness	Deductive
Islands	2D-Grid	Coordinates & Geometry	Spatial Awareness	Abductive
Ordering Text	Multiple words	Strings & Sorting	Arithmetic, Comparative	Deductive

Table 1: Detailed information on TEXTGAMES puzzle games, encompassing a broad spectrum of output formats, categories, skillsets, and reasoning types.

Game	Easy	Medium	Hard
Anagram Scribble	- 3 to 5 letter English word - Character list ≤ 10 - Repeatable use of char	- 6 to 7 letter English word - Character list ≤ 10 - Repeatable use of char	- 8 to 10 letter English word - Character list ≤ 10 - Each char can only be used once
Password	- 2 Rules	- 4 Rules	- 6 Rules
Bracket Game	- Rules = 3 - Words = 3 - Depth = 2	- Rules = 5 - Words = 5 - Depth = 2	- Rules = 5 - Words = 5 - Depth = 3
String Search	- Text length ≤ 10 characters - At most 2 constraints - Multiple solutions may exist - No complex rules	- Text length ≤ 20 characters - At most 3 constraints - Multiple solutions may exist - No complex rules	- Text length ≤ 40 characters - At most 5 constraints - Single solution
Crossword Arranger	- Board size = 3x3 - Words = 8 - 25% Noise words	- Board size = 4x4 - Words = 16 - 50% Noise words	- Board size = 5x5 - Words = 20 - 50% Noise words
Text Sudoku	- Board size = 4x4 - Empty ratio = 0.25	- Board size = 4x4 - Empty Ratio = 0.5	- Board size = 9x9 - Empty Ratio = 0.4
Islands	- Only 1 island - No coconut tree - No complex constraints	- 1 to 3 islands - No complex constraints	- 3 to 6 islands
Ordering Text	- Rules = 2 - Words = 3 - $3 \leq \text{Word Length} \leq 8$	- $2 \leq \text{Rules} \leq 4$ - $4 \leq \text{Words} \leq 6$ - $3 \leq \text{Word Length} \leq 8$	- $4 \leq \text{Rules} \leq 8$ - $6 \leq \text{Words} \leq 10$ - $3 \leq \text{Word Length} \leq 15$

Table 2: Difficulty levels of TEXTGAMES puzzle games detailed with associated constraints and rules.

required, please refer to Table 1. Each game is designed with three levels of difficulty: Easy, Medium, and Hard, with specifics available in Table 2. The difficulty escalates through factors like the increased size of a 2D board, more stringent constraints, and progressively challenging reasoning tasks. Most games are designed to support multiple solutions, which can vary with the difficulty level. For instance, in Anagram Scribble (2.1.1), the same set of characters can be rearranged to create different English words, such as “game” and “mega.” In Islands (2.1.7), the location of coconut trees can be arbitrary. In contrast, Ordering Text (2.1.8) offers only one possible solution, as words with the same score are sorted lexicographically.

2.3 Game Categories

The benchmark tasks can be divided into two categories: 1D and 2D formats. The 1D puzzles include Anagram Scribble (see Section 2.1.1), Password Game (see Section 2.1.2), Bracket Game (see Section 2.1.3), and String Search (see Section 2.1.4). In contrast, the 2D puzzles demand spatial awareness and the capacity to track values across multiple rows. These include Crossword Arranger (see Section 2.1.5), Text Sudoku (see Section 2.1.6), Islands (see Section 2.1.7), and Ordering Text (see Section 2.1.8). Generally, models demonstrate superior performance on 1D puzzles. For instance, the performance of LLMs on easy 2D puzzles is comparable to their performance on medium-difficulty 1D

puzzles, while their performance on medium 2D puzzles parallels that on hard 1D puzzles. This is illustrated in Figure 1, highlighting the challenges LLMs face with 2D spatial reasoning.

2.4 Game Generation

For each game, we create instances by randomly sampling according to the specified rules for each difficulty level, resulting in 1,000 test samples per difficulty. This amounts to a total of 24,000 test samples across all games and difficulty levels. Additionally, we generate a number of training samples for few-shot learning across all difficulties, ensuring that these samples do not overlap with the test set. We refer to the test samples as $\mathcal{D}^{\text{test}}$ and the training samples as $\mathcal{D}^{\text{train}}$.

3 TEXTGAMES Evaluation

For our TEXTGAMES, we design a game evaluation framework where LLMs emulate player behavior to play the games. This system uses a LLM to generate solutions and integrates a grader to verify their correctness. To further test models' performance, we implement multi-turn prompting, enabling the model to iteratively refine its responses. This iterative process involves receiving feedback from the grader, which allows the models to self reflect and attempt to correct the answers.

3.1 Prompt Generation

We utilize in-context learning prompts to generate answers and evaluate the capabilities of LLMs under two configurations: zero-shot and one-shot prompts. Our prompt is defined as $P \leftarrow (T, C, E, I)$, where it is constructed using a prompt template T , along with constraints C , one-shot examples E , and relevant context I from previous interactions for multi-turn scenarios. We denote the LLMs used for inference as θ and the grader that evaluates the correctness of the answers as $\mathcal{G}$. Detailed information about the prompts for each game is provided in Appendix H.

3.2 Multi-Turn Prompting

Algorithm 1 outlines the procedure for implementing multi-turn prompting, a strategy that iteratively refines responses based on feedback from a grader. At each turn, the model generates a response given the test constraint, few-shot examples, and previous interactions. The grader evaluates the response and provides feedback if errors are detected. The interaction history is updated with both the response and

Algorithm 1 TEXTGAMES Evaluation System

Require: LLM θ , Grader $\mathcal{G}$, Template $\mathcal{T}$, Dataset $\mathcal{D}^{\{\text{train}, \text{test}\}}$.
Initialize: Few-shot example(s) $E \subseteq \mathcal{D}^{\text{train}}$.
Initialize: Maximum Turn $N = 3$.
1: **for all** Constraints $C \in \mathcal{D}^{\text{test}}$ **do**
2: $I \leftarrow []$
3: **for** $i = 1, \dots, N$ **do**
4: $P \leftarrow (T, C, E, I)$ $\triangleright$ Prompt construction
5: $R \leftarrow \theta(P)$ $\triangleright$ LLM Response
6: $S, F \leftarrow \mathcal{G}(C, R)$ $\triangleright$ is_solve, feedback
7: **if** S is **True** **then**
8: Break the for loop
9: **else**
10: $I \leftarrow I + [R, F]$ $\triangleright$ Update interactions
11: **end if**
12: **end for**
13: **end for**

feedback, allowing the model to adjust its outputs in subsequent turns. The process terminates early if the grader confirms a correct response, ensuring adaptability while enabling iterative refinement. A complete list of feedback for all games can be found in Appendix J.

4 Experimental Setup

For each task described in Section 2, we begin by developing a grader to verify the correctness of the answers. These graders function similarly to those used on online judge platforms or in competitive programming contests, focusing solely on determining whether an answer is correct or incorrect. Subsequently, we evaluate the performance of various LLMs using these graders. Additionally, we have created a web-based platform to collect data for testing human performance on the same tasks, allowing for a comprehensive comparison between human and model capabilities.

4.1 Models

We employ several open-sourced LLMs known for their competitive performance on various benchmarks, including Gemma-2 9B and 27B Instruct (Team et al., 2024), Llama-3.1 8B Instruct, Llama-3.3 70B Instruct (Dubey et al., 2024), and the Qwen-2.5 instruct models of different scales (7B, 14B, 32B, and 72B) (Yang et al., 2024a). Additionally, we include proprietary closed models like GPT-4o Mini and GPT-3o Mini, given that mini models offer a good balance between performance and cost efficiency. For model inference, we implement greedy decoding to maintain deterministic outcomes. Specifically, for GPT-o3 Mini, we configure the settings to prioritize the shortest reasoning generation option. We use accuracy or

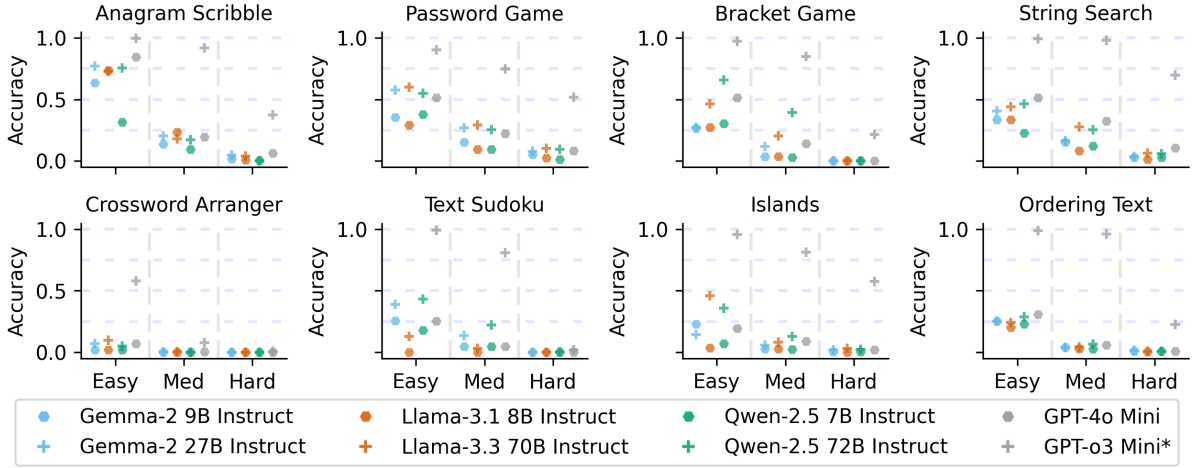


Figure 3: LLM Results on TEXTGAMES BENCHMARK in the one-shot setting. Med indicates Medium-difficulty level. *For GPT-o3 Mini, we present the results from zero-shot setting.

solve rate as our evaluation metric to measure the correctness of the answer.

4.2 Human Annotation

To understand how humans play and to compare their abilities with those of LLMs, we develop a web-based interface² that enables human participants to engage with our games. Through this platform, we document interactions between participants and our grading system, capturing metrics such as solve rates, the number of attempts, and the time taken to solve. These data allow us to directly compare human capabilities to those of LLMs. Each participant is asked to solve 2 to 3 different sessions. Details regarding the demographics of the annotators are available in Appendix E.

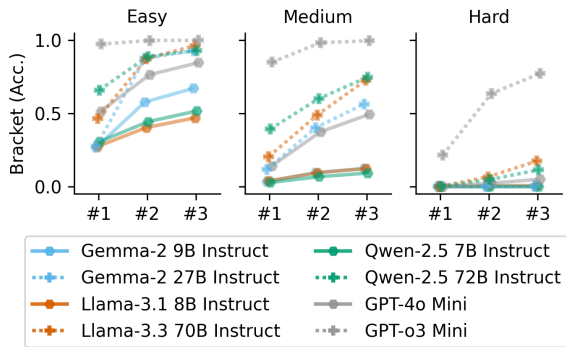


Figure 4: LLM performance on the Bracket Game in the one-shot setting, excluding GPT models. The results show that increasing the number of turns generally enhances performance. A similar trend is evident in Crossword Arranger, as shown by Figure 7 in the Appendix F showing illustrations from all games

²The link will be released after acceptance.

5 Results and Analysis

Our findings indicate that our benchmark poses a considerable challenge for LLMs as illustrated in Figure 3. Even at the easiest difficulty level, the majority of models struggle to solve the games. An exception is the highly capable GPT-o3 Mini, which succeeds on only a subset of the games. This highlights the persistent difficulty of our benchmark for LLMs, highlighting areas where further advancements are needed.

Model Scaling Improves Performance. Larger models generally exhibit superior performance, particularly when comparing models within the same family (e.g., Gemma-2 9B vs. 27B Instruct), where the larger model consistently outperforms its smaller counterpart. Notably, the Gemma-2 27B Instruct model remains highly competitive despite being significantly smaller than other 70B+ baselines. Typically, larger models excel on easier tasks; however, this advantage does not necessarily extend to more challenging tasks, such as those requiring reasoning in two-dimensional coordinates. This trend is illustrated in Figure 6 in the Appendix.

Multi-Turn Feedback Improves LLM Performance. While LLMs typically underperform on single-turn attempts, we observe noteworthy improvements when they receive feedback explaining why their previous responses were incorrect. These enhancements are most evident at the easy difficulty level. Figures 4 and 7 illustrate this positive trend, showcasing how LLMs effectively use feedback from previous interactions to engage in self-reflection and refine their subsequent outputs.

Model	Turn #	Easy			Medium			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		39.8	58.0	64.1	12.0	19.7	25.2	2.5	3.6	4.2
Gemma-2 27B Instruct		50.7	77.0	82.4	18.9	37.5	46.7	4.1	7.1	9.3
Llama-3.1 8B Instruct		40.8	52.4	58.6	11.2	16.2	18.9	1.2	2.1	2.6
Llama-3.3 70B Instruct		55.8	<u>78.7</u>	<u>86.4</u>	23.9	<u>43.0</u>	<u>56.6</u>	5.1	<u>10.2</u>	<u>15.3</u>
Qwen-2.5 7B Instruct		30.6	44.6	52.5	8.4	14.2	18.5	1.2	1.8	2.3
Qwen-2.5 72B Instruct		<u>60.7</u>	75.4	81.5	<u>26.9</u>	40.3	49.3	3.9	7.8	11.1
GPT-4o Mini		59.6	74.3	79.0	22.1	37.6	45.3	<u>6.3</u>	9.4	11.6
GPT-o3 Mini		96.5	98.9	99.4	87.2	96.2	97.4	45.1	69.5	78.0

Table 3: Average solve rate (%) for multi-turn 1D Puzzles.

Model	Turn #	Easy			Medium			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		19.1	30.4	36.7	2.9	5.1	7.9	0.3	0.8	1.6
Gemma-2 27B Instruct		21.4	36.0	44.4	5.9	9.3	13.5	0.9	1.6	2.6
Llama-3.1 8B Instruct		6.6	12.7	20.7	1.4	2.0	3.7	0.1	0.3	0.7
Llama-3.3 70B Instruct		23.2	38.0	48.4	4.0	7.3	10.7	<u>0.9</u>	<u>2.2</u>	3.4
Qwen-2.5 7B Instruct		12.5	20.7	26.3	2.6	4.8	6.5	0.3	0.7	1.3
Qwen-2.5 72B Instruct		<u>28.2</u>	<u>43.1</u>	<u>51.1</u>	<u>10.4</u>	<u>15.7</u>	<u>19.1</u>	0.9	2.1	<u>3.4</u>
GPT-4o Mini		20.7	34.7	40.9	5.1	9.5	12.4	0.7	1.8	2.6
GPT-o3 Mini		87.8	94.9	98.8	66.5	80.7	86.9	20.6	40.5	48.6

Table 4: Average solve rate (%) for multi-turn 2D Puzzles.

	1 st Turn Solve Rate (%)			Avg. Attempts			Avg. Time to Solve (s)		
	Easy	Medium	Hard	Easy	Medium	Hard	Easy	Medium	Hard
1D Puzzles									
Anagram Scribble	100.0	87.5	57.1	1.00	1.12	2.14	11.7	82.6	263.5
Password Game	88.9	100.0	44.4	1.22	1.00	1.78	27.2	44.4	73.4
Bracket Game	100.0	75.0	75.0	1.00	1.25	1.25	29.3	48.9	71.2
String Search	100.0	100.0	75.0	1.00	1.00	1.38	14.6	17.4	41.4
2D Puzzles									
Crossword Arranger	77.8	100.0	88.9	1.33	1.00	1.11	32.2	138.7	128.2
Text Sudoku	100.0	100.0	77.8	1.00	1.00	1.78	11.7	29.5	536.3
Islands	100.0	100.0	100.0	1.00	1.00	1.00	12.2	25.5	41.4
Ordering Text	55.6	57.1	42.9	1.67	3.14	2.00	72.3	127.5	424.3

Table 5: Performance of human annotators on playing TEXTGAMES BENCHMARK.

A similar trend is evident in the results for various models, as shown in Table 3 for 1D games and Table 4 for 2D games.

TEXTGAMES Are Solvable by Humans. When comparing LLM performance to human performance, we observe that humans can easily achieve full scores, especially on the easy difficulty. This is because some problems, particularly at lower difficulty levels, are arguably trivial for adult humans. On average, humans could solve all the problems within 2 attempts except for Ordering Text on the medium difficulty. This finding is particu-

larly interesting given that recent research suggests LLMs exhibit intelligence seemingly on par with humans (Achiam et al., 2023). Yet, these models struggle with tasks as simple as searching for a substring and placing a bracket around it or constructing a 2D string with a predefined number of “islands.” At higher difficulty levels, we observe a decline in human performance, reflected in the lower one-turn solve rate and increased time required to solve. However, while LLMs exhibit a similar trend, most models fail to solve any hard problems, whereas humans still manage to solve

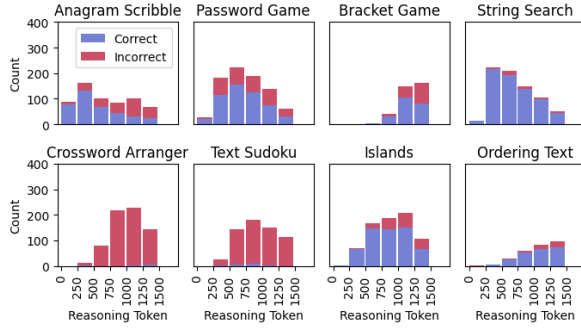


Figure 5: In hard games, the test-time scaling of GPT-o3 Mini displays inverse scaling behavior, with longer reasoning traces often leading to incorrect results.

them in one turn.

Misaligned Difficulty Perception between LLMs and Humans. The “Islands” and “String Search” games are among the easiest problems for humans; even at the hardest difficulty, humans typically solve them in fewer than two turns, making them some of the fastest problems to complete. In contrast, LLMs struggle significantly with these tasks, generally exhibiting subpar performance. This highlights a discrepancy in difficulty perception between humans and LLMs and sheds light on the fundamental differences in how humans and LLMs approach constrained puzzle-solving.

Inverse-Scaling on Reasoning Length and Performance Previous studies have generally shown that longer reasoning sequences enhance performance. Interestingly, this pattern is not evident in GPT-o3 Mini (Figure 5). We observe that GPT-o3 Mini tends to produce incorrect answers more frequently with extended reasoning tokens, particularly in the Bracket Game, Islands, and Ordering Text. Although GPT-o3 Mini does not disclose its reasoning process, we hypothesize that it may become confused by its own extended reasoning, resulting in overcomplicated solutions or incorrect understanding. An empirical example is illustrated by the recent DeepSeek R1 hallucination, where the system initially provided a correct answer but, after further analysis and reasoning, can be misled into an incorrect conclusion, shown in Table 26 in the Appendix I.

6 Related Work

Games using LLMs. With the advancement of LLMs, recent works examine their capabilities in playing games or assisting humans in game-

play (Hu et al., 2024a). Classical games like Go (Silver et al., 2017), chess (Feng et al., 2024), Poker (Huang et al., 2024a) have been used as initial testbeds for evaluating models’ planning and decision-making abilities. More recently, more works have explored other genres for more dynamic and complex situations like text-based games (Xiao and Yang, 2024; Stojanovski, 2024; Kazemi et al., 2024), communication games (Guan et al., 2025; Xu et al., 2025), and modern strategic video games (Zhang et al., 2023; Hu et al., 2024b; Qi et al., 2024; Rao et al., 2024; Ma et al., 2025). In comparison, TEXTGAMES takes inspiration from real-life text puzzle games and emphasizes evaluating LLM’s capabilities in simple logic reasoning. Additionally, each game come with different level of difficulty for assessing the models’ robustness.

Text-based Reasoning. Text-based reasoning has been extensively studied across various domains, including commonsense reasoning (Rajani et al., 2019; Bhargava and Ng, 2022; Zhao et al., 2023), mathematical reasoning (Patel et al., 2021; Zhao et al., 2022; Lu et al., 2023), logical reasoning (Pan et al., 2023), causal reasoning (Wang, 2024; Jin et al., 2024), and agent-based reasoning (Motwani et al., 2024). While existing benchmarks assess different aspects of reasoning, they often evaluate these abilities in isolation. In contrast, TEXTGAMES assesses LLMs’ capacity for integrating multiple reasoning skills, offering a richer evaluation of model strengths and weaknesses.

7 Conclusion

We present TEXTGAMES, a text-based puzzle game benchmark designed to evaluate the diverse reasoning abilities of LLMs, including pattern recognition, spatial awareness, arithmetic, and logical reasoning. In addition to only evaluating single-turn solve rate, our evaluation system also implement feedback in multi-turn gameplay settings and test whether models improve through self-reflection. Results show that while LLMs proficiently solve most easy and medium-level problems, they encounter significant challenges with more difficult tasks that demand comprehensive reasoning. In contrast, humans can solve all tasks given sufficient time. We show significant performance improvement with multi-turn prediction via self-reflection. We hope TEXTGAMES could contribute to uncovering and analyzing the weaknesses of LLMs in complex reasoning tasks.

Limitations

In this paper, we focus our investigation by not exhaustively evaluating every possible model, owing to resource constraints. Instead, our primary objective is to develop a benchmark that serves as a platform for future research exploration on reasoning.

Ethical Considerations

In conducting our research, which focuses on evaluating LLMs for complex reasoning tasks, we are committed to upholding the highest standards of transparency and fairness in all aspects of our data collection and evaluation processes. We ensure that the methodologies and criteria used for assessment are clearly documented and unbiased, promoting fair comparisons across different models. Our commitment to these principles aims to foster trust and accountability in our research outcomes.

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A GPU computation usage and Hyperparameters	772
We employ NVIDIA GPUs, RTX A6000 (48GB) and RTX 6000 (48GB), to run inference for the whole open model which took us the equivalent of ~650 GPU hours. We apply the default parameters as defined from each models respective Hugging-Face’s page for all of our experiments. To allow reproducibility, we use greedy decoding, i.e. by setting parameter do_sample to False.	773
B Dataset License	774
We will release our dataset under the open-source CC-BY-SA 4.0 license, facilitating redistribution for future research.	775
C Attribution	776
The icon images on Figure 2 is taken from https://flaticon.com . They are freely for personal and commercial use with attribution.	777

D Model Scale Improvement

Figure 6 illustrates how the scale of the model impacts performance, with variations depending on task difficulty.

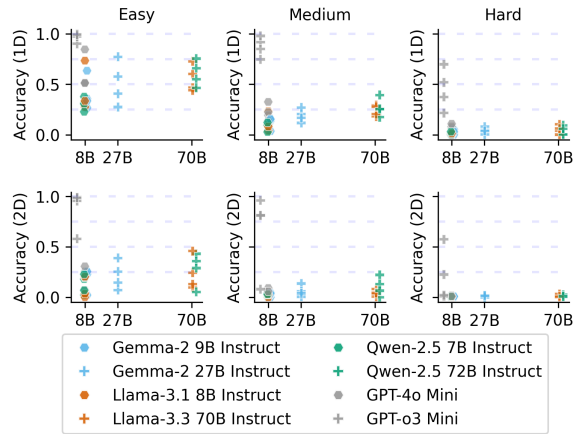


Figure 6: Model scaling improves easier tasks.

E Annotator Demographic

There are 4 annotators, within the age range of 25-35 years old, voluntarily participating in our experiments. All annotators are from Computer Science background with a degree of magisterial or doctoral. All 4 annotators are fluent English speakers from Asia-based origins with experience living in English-speaking countries and have been using English for more than 15 years. All annotators have given consent for using, releasing and redistributing their annotations.

F Multi-turn Results Visualization

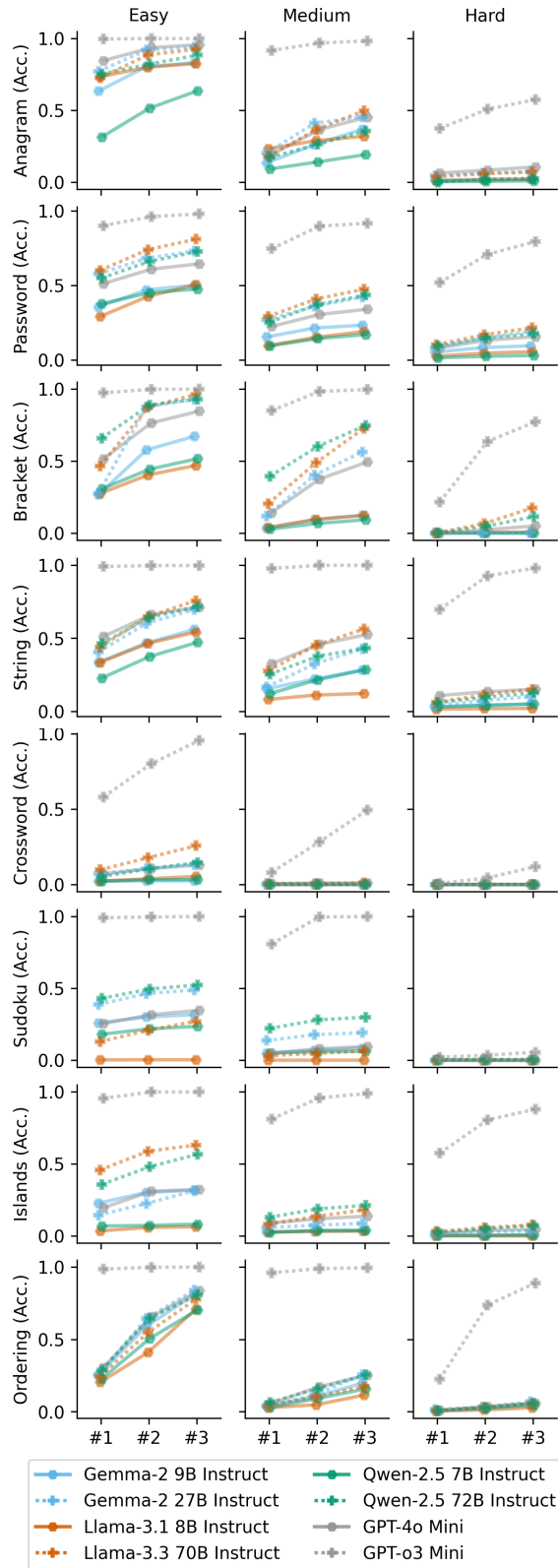


Figure 7: Multi-turn with feedback based on the performance for each game puzzle.

G Complete Experiment Results

We report the complete results that include all the models we evaluated on, as illustrated in Figure 8. The numerical results of these models can be found in Table 6 and Table 7, with the respective Zero-Shot setting performance in Table 8 and Table 9. We also report the performance of multi-turn settings for each game: Anagram Scribble in Table 10, Password Games in Table 11, Bracket Game in Table 12, String Search in Table 13, Crossword Arranger in Table 14, Text Sudoku in Table 15, Islands in Table 16, and Ordering Text in Table 17. †Indicates first 20% of dataset only.

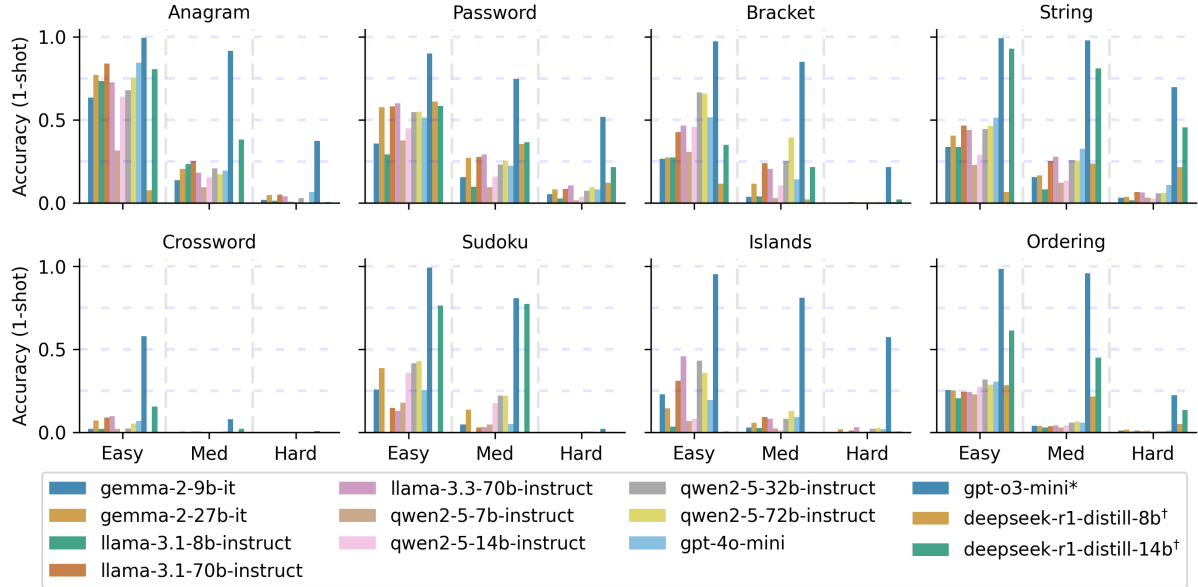


Figure 8: Complete LLMs results against TEXTGAMES BENCHMARK. *We present zero-shot results as reference

	Anagram			Password			Bracket			String		
	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard
Gemma-2 9B Instruct	63.4	13.6	1.6	35.6	15.4	5.2	26.6	3.5	0.0	33.5	15.4	3.0
Gemma-2 27B Instruct	77.1	20.4	4.7	57.7	26.9	8.0	27.4	11.6	0.1	40.6	16.6	3.6
Llama-3.1 8B Instruct	73.4	23.4	1.1	29.1	9.6	2.5	27.4	3.8	0.0	33.5	8.1	1.4
Llama-3.1 70B Instruct	84.0	25.1	5.0	58.1	27.5	8.3	42.7	23.9	0.3	46.6	25.3	6.5
Llama-3.3 70B Instruct	72.7	18.1	3.9	60.1	29.2	10.3	46.5	20.4	0.0	44.0	27.8	6.2
Qwen-2.5 7B Instruct	31.5	9.3	0.6	37.7	9.4	1.4	30.6	2.7	0.0	22.7	12.1	3.0
Qwen-2.5 14B Instruct	64.0	15.4	0.5	44.9	15.8	3.7	45.7	10.4	0.0	28.8	13.3	2.6
Qwen-2.5 32B Instruct	67.9	20.8	2.9	54.8	23.1	7.3	<u>66.5</u>	25.1	0.1	44.4	25.7	5.8
Qwen-2.5 72B Instruct	75.4	17.4	0.2	55.0	25.5	9.3	65.9	<u>39.3</u>	0.3	46.4	25.4	5.9
GPT-4o Mini	<u>84.5</u>	19.4	<u>6.5</u>	51.2	22.4	8.1	51.5	14.1	0.0	51.2	32.5	10.8
GPT-o3 Mini*	99.6	91.6	37.4	90.1	74.6	51.9	97.3	84.9	21.5	99.2	97.9	69.8
DeepSeek-R1-Distill 8B†	7.5	0.0	0.0	<u>61.0</u>	35.5	12.0	11.5	2.0	0.5	6.5	23.5	21.5
DeepSeek-R1-Distill 14B†	80.5	<u>38.0</u>	0.5	58.5	<u>36.5</u>	<u>21.5</u>	35.0	21.5	<u>2.0</u>	<u>93.0</u>	<u>81.0</u>	<u>45.5</u>

Table 6: Complete Average Results (%) for 1D Puzzles. *Zero-shot results as reference

	Crossword			Sudoku			Islands			Ordering		
	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard
Gemma-2 9B Instruct	2.1	0.0	0.0	25.9	4.8	0.0	22.8	2.9	0.3	25.5	4.0	0.9
Gemma-2 27B Instruct	7.1	0.5	0.0	38.8	13.6	0.0	14.5	5.8	1.9	25.4	3.8	1.5
Llama-3.1 8B Instruct	2.2	0.0	0.0	0.1	0.0	0.0	3.5	2.7	0.1	20.4	2.8	0.5
Llama-3.1 70B Instruct	8.9	0.4	<u>0.1</u>	14.7	2.8	0.0	31.2	9.1	1.0	24.5	3.6	1.0
Llama-3.3 70B Instruct	9.7	0.6	0.0	12.9	3.1	0.0	<u>45.8</u>	8.1	<u>3.1</u>	24.3	4.2	0.7
Qwen-2.5 7B Instruct	2.2	0.2	0.0	18.0	4.8	0.0	6.9	2.4	0.3	22.8	3.0	0.9
Qwen-2.5 14B Instruct	0.0	0.0	0.0	35.9	17.7	0.3	8.2	1.2	0.5	27.4	4.1	0.5
Qwen-2.5 32B Instruct	2.3	0.0	0.0	41.7	22.0	<u>0.4</u>	43.3	8.1	2.2	31.9	5.7	0.4
Qwen-2.5 72B Instruct	5.2	0.0	0.0	43.0	22.1	<u>0.4</u>	35.8	<u>13.0</u>	2.5	28.8	6.5	0.8
GPT-4o Mini	6.9	0.4	0.0	25.5	4.9	0.0	19.6	9.1	1.9	30.6	5.8	0.9
GPT-o3 Mini*	57.9	8.0	0.7	99.2	80.9	2.0	95.5	81.1	57.4	98.6	96.0	22.5
DeepSeek-R1-Distill 8B [†]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	28.5	21.5	5.0
DeepSeek-R1-Distill 14B [†]	<u>15.5</u>	<u>2.0</u>	0.0	<u>76.5</u>	<u>77.5</u>	0.0	0.5	0.0	0.5	<u>61.5</u>	<u>45.0</u>	<u>13.5</u>

Table 7: Complete Average Results (%) for 2D Puzzles. *Zero-shot results as reference

	Anagram			Password			Bracket			String		
	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard
Gemma-2 9B Instruct	77.7	14.2	1.0	36.4	13.8	4.6	2.3	0.3	0.0	37.7	20.2	5.3
Gemma-2 27B Instruct	<u>88.0</u>	31.8	3.1	45.9	18.7	6.6	29.8	5.4	0.0	44.1	23.7	4.4
Llama-3.1 8B Instruct	56.3	7.5	0.1	31.8	7.2	1.3	0.1	0.0	0.0	17.2	6.9	0.3
Llama-3.1 70B Instruct	69.5	27.7	2.5	45.1	19.0	5.6	23.0	6.0	0.4	45.9	25.4	5.8
Llama-3.3 70B Instruct	77.4	30.9	3.2	47.5	20.0	5.9	34.0	14.0	0.6	45.0	26.3	5.4
Qwen-2.5 7B Instruct	8.2	0.2	0.0	32.9	9.2	1.3	6.4	1.1	0.0	25.4	11.1	1.4
Qwen-2.5 14B Instruct	23.1	7.2	0.6	34.0	12.0	2.4	22.9	3.7	0.0	32.7	14.4	1.8
Qwen-2.5 32B Instruct	68.7	16.7	2.3	47.2	20.2	6.0	47.6	15.8	0.3	41.4	27.0	6.1
Qwen-2.5 72B Instruct	30.0	0.4	0.1	50.0	21.8	8.3	<u>58.3</u>	19.9	0.0	46.7	26.7	7.0
GPT-4o Mini	79.9	26.7	<u>5.7</u>	46.9	18.9	6.5	26.6	7.2	0.0	39.8	28.1	7.6
GPT-o3 Mini	99.6	91.6	37.4	90.1	74.6	51.9	97.3	84.9	21.5	99.2	97.9	69.8
DeepSeek-R1-Distill 8B [†]	7.5	0.0	0.0	61.0	35.5	12.0	11.5	2.0	0.5	6.5	23.5	21.5
DeepSeek-R1-Distill 14B [†]	81.0	<u>35.0</u>	2.0	<u>67.0</u>	<u>45.5</u>	<u>16.5</u>	51.5	<u>38.5</u>	<u>5.0</u>	<u>94.5</u>	<u>68.5</u>	<u>58.0</u>

Table 8: Complete Average Results (%) for 1D Puzzles (Zero-Shot).

	Crossword			Sudoku			Islands			Ordering		
	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard	Easy	Med	Hard
Gemma-2 9B Instruct	1.1	0.0	0.0	24.3	3.1	0.0	1.8	0.0	0.5	20.6	2.6	0.9
Gemma-2 27B Instruct	6.6	0.0	0.0	39.1	15.3	0.0	10.3	2.4	0.0	21.9	2.9	1.2
Llama-3.1 8B Instruct	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.3	0.0	0.3	0.0	0.0
Llama-3.1 70B Instruct	12.2	0.0	0.0	7.2	2.7	0.0	0.0	0.0	0.0	8.4	0.9	0.4
Llama-3.3 70B Instruct	5.5	0.1	0.0	7.1	1.4	0.0	1.9	3.2	0.8	2.1	0.1	0.0
Qwen-2.5 7B Instruct	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	22.1	2.0	0.3
Qwen-2.5 14B Instruct	1.3	0.0	0.0	31.1	15.1	0.3	0.8	1.3	0.3	18.6	2.7	0.8
Qwen-2.5 32B Instruct	7.3	0.0	0.0	34.2	15.5	0.5	0.0	0.0	0.0	27.3	5.2	0.7
Qwen-2.5 72B Instruct	0.0	0.0	0.0	<u>42.7</u>	20.0	0.1	0.0	0.0	0.0	22.9	4.6	0.6
GPT-4o Mini	14.0	4.6	<u>0.1</u>	1.1	0.2	0.0	<u>31.8</u>	5.2	0.8	22.6	2.5	1.0
GPT-o3 Mini	57.9	8.0	0.7	99.2	80.9	2.0	95.5	81.1	57.4	98.6	96.0	22.5
DeepSeek-R1-Distill 8B [†]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	28.5	21.5	5.0
DeepSeek-R1-Distill 14B [†]	<u>15.5</u>	4.0	0.0	34.0	<u>22.0</u>	<u>1.0</u>	3.0	<u>10.0</u>	<u>8.0</u>	<u>65.0</u>	<u>59.0</u>	<u>16.5</u>

Table 9: Complete Average Results (%) for 2D Puzzles (Zero-Shot).

Model	Turn #	Anagram								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		63.4	80.4	83.1	13.6	26.2	36.9	1.6	2.0	2.3
Gemma-2 27B Instruct		77.1	91.9	94.0	20.4	<u>41.1</u>	45.6	4.7	6.4	7.5
Llama-3.1 8B Instruct		73.4	80.0	82.6	23.4	29.0	32.0	1.1	1.7	2.3
Llama-3.1 70B Instruct		84.0	91.1	93.4	<u>25.1</u>	40.6	49.3	5.0	7.0	7.8
Llama-3.3 70B Instruct		72.7	88.6	92.5	18.1	36.7	<u>49.8</u>	3.9	6.0	7.2
Qwen-2.5 7B Instruct		31.5	51.6	63.6	9.3	14.2	19.3	0.6	0.7	1.1
Qwen-2.5 14B Instruct		64.0	78.0	83.9	15.4	21.1	26.3	0.5	1.3	1.9
Qwen-2.5 32B Instruct		67.9	83.6	88.4	20.8	32.9	42.5	2.9	3.5	4.0
Qwen-2.5 72B Instruct		75.4	82.2	88.4	17.4	26.4	35.6	0.2	2.2	2.8
GPT-4o Mini		<u>84.5</u>	<u>93.6</u>	<u>95.6</u>	19.4	36.7	45.3	<u>6.5</u>	<u>8.5</u>	<u>10.7</u>
GPT-o3 Mini		99.6	99.9	99.9	91.6	96.8	98.3	37.4	50.8	57.5

Table 10: 3-Turns Accuracy (%) of Anagram Scribble.

Model	Turn #	Password								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		35.6	47.0	50.1	15.4	21.4	23.3	5.2	8.2	9.5
Gemma-2 27B Instruct		57.7	68.2	73.1	26.9	36.0	42.1	8.0	14.6	19.3
Llama-3.1 8B Instruct		29.1	42.6	50.5	9.6	15.2	19.0	2.5	4.5	5.5
Llama-3.1 70B Instruct		58.1	73.2	79.4	27.5	40.8	<u>47.7</u>	8.3	15.7	21.5
Llama-3.3 70B Instruct		<u>60.1</u>	<u>74.0</u>	<u>81.0</u>	<u>29.2</u>	<u>40.9</u>	47.4	<u>10.3</u>	<u>16.9</u>	<u>21.6</u>
Qwen-2.5 7B Instruct		37.7	45.1	47.6	9.4	14.3	17.0	1.4	2.4	3.0
Qwen-2.5 14B Instruct		44.9	61.8	67.2	15.8	26.4	32.7	3.7	7.6	9.6
Qwen-2.5 32B Instruct		54.8	68.7	74.4	23.1	36.3	43.3	7.3	14.5	18.2
Qwen-2.5 72B Instruct		55.0	66.1	72.7	25.5	37.2	43.5	9.3	14.8	17.6
GPT-4o Mini		51.2	60.9	64.5	22.4	30.5	34.0	8.1	13.3	15.6
GPT-o3 Mini		90.1	96.2	98.0	74.6	89.8	91.7	51.9	70.9	79.4

Table 11: 3-Turns Accuracy (%) of Password Game.

Model	Turn #	Bracket								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		26.6	57.7	67.1	3.5	9.3	12.4	0.0	0.0	0.0
Gemma-2 27B Instruct		27.4	87.5	92.9	11.6	40.2	56.4	0.1	0.2	0.4
Llama-3.1 8B Instruct		27.4	40.3	46.9	3.8	9.5	12.2	0.0	0.5	0.7
Llama-3.1 70B Instruct		42.7	81.6	94.3	23.9	43.7	68.9	<u>0.3</u>	4.3	<u>19.2</u>
Llama-3.3 70B Instruct		46.5	87.3	<u>96.3</u>	20.4	48.9	72.7	0.0	<u>6.6</u>	17.4
Qwen-2.5 7B Instruct		30.6	44.3	51.6	2.7	6.7	9.2	0.0	0.0	0.0
Qwen-2.5 14B Instruct		45.7	61.3	69.9	10.4	16.1	23.9	0.0	0.1	0.8
Qwen-2.5 32B Instruct		<u>66.5</u>	82.1	87.1	25.1	41.4	50.1	0.1	2.8	6.2
Qwen-2.5 72B Instruct		65.9	<u>88.9</u>	92.9	<u>39.3</u>	<u>60.1</u>	<u>74.7</u>	<u>0.3</u>	4.7	11.4
GPT-4o Mini		51.5	76.4	84.6	14.1	37.4	49.3	0.0	2.3	5.0
GPT-o3 Mini		97.3	99.8	99.9	84.9	98.3	99.6	21.5	63.6	77.2

Table 12: 3-Turns Accuracy (%) of Bracket Game.

Model	Turn #	String Search								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		33.5	46.8	56.0	15.4	22.0	28.4	3.0	4.2	4.9
Gemma-2 27B Instruct		40.6	60.2	69.7	16.6	32.5	42.7	3.6	7.2	10.1
Llama-3.1 8B Instruct		33.5	46.5	54.2	8.1	11.1	12.2	1.4	1.8	1.9
Llama-3.1 70B Instruct		46.6	66.0	74.5	25.3	41.2	49.2	6.5	10.9	14.1
Llama-3.3 70B Instruct		44.0	64.9	<u>75.8</u>	27.8	45.3	<u>56.4</u>	6.2	11.3	15.0
Qwen-2.5 7B Instruct		22.7	37.5	47.2	12.1	21.6	28.5	3.0	4.0	5.0
Qwen-2.5 14B Instruct		28.8	48.5	59.0	13.3	23.7	31.7	2.6	3.5	4.5
Qwen-2.5 32B Instruct		44.4	61.4	70.0	25.7	41.0	50.4	5.8	10.0	12.8
Qwen-2.5 72B Instruct		46.4	64.3	71.8	25.4	37.7	43.3	5.9	9.6	12.6
GPT-4o Mini		<u>51.2</u>	<u>66.2</u>	71.4	<u>32.5</u>	<u>45.9</u>	52.6	<u>10.8</u>	<u>13.4</u>	<u>15.1</u>
GPT-o3 Mini		99.2	99.8	99.9	97.9	100.0	100.0	69.8	92.8	98.1

Table 13: 3-Turns Accuracy (%) of String Search.

Model	Turn #	Crossword								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		2.1	2.4	2.4	0.0	0.0	0.0	0.0	0.0	0.0
Gemma-2 27B Instruct		7.1	10.3	13.2	0.5	0.5	0.5	0.0	0.0	0.0
Llama-3.1 8B Instruct		2.2	3.7	5.3	0.0	0.0	0.0	0.0	0.0	0.0
Llama-3.1 70B Instruct		8.9	17.5	<u>26.0</u>	0.4	0.5	0.8	<u>0.1</u>	<u>0.1</u>	<u>0.1</u>
Llama-3.3 70B Instruct		<u>9.7</u>	<u>17.7</u>	25.6	<u>0.6</u>	0.6	<u>1.0</u>	0.0	0.0	0.0
Qwen-2.5 7B Instruct		2.2	3.1	3.2	0.2	0.2	0.2	0.0	0.0	0.0
Qwen-2.5 14B Instruct		0.0	0.6	1.4	0.0	0.0	0.0	0.0	0.0	0.0
Qwen-2.5 32B Instruct		2.3	4.2	5.1	0.0	0.0	0.0	0.0	0.0	0.0
Qwen-2.5 72B Instruct		5.2	10.4	14.6	0.0	0.0	0.0	0.0	0.0	0.0
GPT-4o Mini		6.9	10.6	12.9	0.4	<u>0.9</u>	<u>1.0</u>	0.0	0.0	0.0
GPT-o3 Mini		57.9	80.0	95.4	8.0	28.2	49.3	0.7	4.3	11.8

Table 14: 3-Turns Accuracy (%) of Crossword Arranger.

Model	Turn #	Sudoku								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		25.9	30.0	31.7	4.8	6.8	8.3	0.0	0.0	0.0
Gemma-2 27B Instruct		38.8	46.3	48.9	13.6	17.7	19.1	0.0	0.0	0.0
Llama-3.1 8B Instruct		0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0
Llama-3.1 70B Instruct		14.7	24.8	29.1	2.8	5.0	6.9	0.0	0.0	0.0
Llama-3.3 70B Instruct		12.9	20.7	27.1	3.1	4.6	6.3	0.0	0.0	0.0
Qwen-2.5 7B Instruct		18.0	21.8	23.5	4.8	5.8	6.2	0.0	0.0	0.0
Qwen-2.5 14B Instruct		35.9	44.1	47.3	17.7	22.1	24.9	0.3	0.3	0.3
Qwen-2.5 32B Instruct		41.7	47.5	49.8	22.0	25.7	28.1	<u>0.4</u>	<u>0.4</u>	<u>0.4</u>
Qwen-2.5 72B Instruct		<u>43.0</u>	<u>49.5</u>	<u>52.3</u>	<u>22.1</u>	<u>28.0</u>	<u>29.9</u>	<u>0.4</u>	<u>0.4</u>	<u>0.4</u>
GPT-4o Mini		25.5	31.6	34.7	4.9	7.9	9.5	0.0	0.0	0.0
GPT-o3 Mini		99.2	99.7	100.0	80.9	99.7	100.0	2.0	3.4	5.5

Table 15: 3-Turns Accuracy (%) of Text Sudoku.

Model	Turn #	Islands								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		22.8	30.2	31.8	2.9	3.1	3.6	0.3	0.4	0.9
Gemma-2 27B Instruct		14.5	22.5	31.4	5.8	7.4	8.7	1.9	2.9	3.5
Llama-3.1 8B Instruct		3.5	5.9	6.4	2.7	3.3	3.4	0.1	0.1	0.1
Llama-3.1 70B Instruct		31.2	35.4	36.4	9.1	14.7	17.0	1.0	3.5	5.2
Llama-3.3 70B Instruct		<u>45.8</u>	<u>58.8</u>	<u>63.1</u>	8.1	13.6	18.1	<u>3.1</u>	<u>5.7</u>	<u>7.7</u>
Qwen-2.5 7B Instruct		6.9	7.3	8.2	2.4	3.9	3.9	0.3	0.4	0.6
Qwen-2.5 14B Instruct		8.2	12.1	14.7	1.2	1.6	2.0	0.5	1.2	2.0
Qwen-2.5 32B Instruct		43.3	56.4	60.6	8.1	11.4	16.1	2.2	3.3	4.9
Qwen-2.5 72B Instruct		35.8	48.0	56.7	<u>13.0</u>	<u>18.8</u>	<u>21.2</u>	2.5	5.1	7.1
GPT-4o Mini		19.6	31.0	32.3	9.1	11.9	13.8	1.9	3.7	4.5
GPT-o3 Mini		95.5	100.0	100.0	81.1	95.9	98.9	57.4	80.5	88.1

Table 16: 3-Turns Accuracy (%) of Islands.

Model	Turn #	Ordering								
		Easy			Med			Hard		
		#1	#2	#3	#1	#2	#3	#1	#2	#3
Gemma-2 9B Instruct		25.5	59.0	81.0	4.0	10.3	19.8	0.9	2.6	5.6
Gemma-2 27B Instruct		25.4	64.7	83.9	3.8	11.6	25.5	<u>1.5</u>	3.3	<u>6.9</u>
Llama-3.1 8B Instruct		20.4	41.0	71.0	2.8	4.8	11.5	0.5	1.2	2.8
Llama-3.1 70B Instruct		24.5	56.4	79.1	3.6	12.3	20.9	1.0	3.4	6.6
Llama-3.3 70B Instruct		24.3	54.8	77.8	4.2	10.4	17.3	0.7	3.1	5.7
Qwen-2.5 7B Instruct		22.8	50.6	70.4	3.0	9.2	15.8	0.9	2.5	4.5
Qwen-2.5 14B Instruct		27.4	63.7	82.3	4.1	14.1	23.1	0.5	2.8	4.8
Qwen-2.5 32B Instruct		<u>31.9</u>	<u>69.3</u>	<u>84.0</u>	5.7	<u>18.5</u>	<u>27.6</u>	0.4	3.1	6.8
Qwen-2.5 72B Instruct		28.8	64.7	80.8	<u>6.5</u>	16.0	25.5	0.8	2.8	6.0
GPT-4o Mini		30.6	65.6	83.8	5.8	17.3	25.4	0.9	<u>3.7</u>	5.9
GPT-o3 Mini		98.6	99.9	100.0	96.0	99.0	99.5	22.5	73.8	89.0

Table 17: 3-Turns Accuracy (%) of Ordering Text.

826
827
828
829

H Prompt Templates and Games Constraints

We detail the prompt templates and constraints for prompt constructions here: Anagram Scribble in Table 18, Password Games in Table 19, Bracket Game in Table 20, String Search in Table 21, Crossword Arranger in Table 22, Text Sudoku in Table 23, Islands in Table 24, and Ordering Text in Table 25.

<Prompt Template ($\mathcal{P}$)> Construct a valid $[N]$-character English word from the following letters: '$[C_1]$', '$[C_2]$', ..., '$[C_{N+m}]$'. Each character can be used multiple times. Please write None if there is no valid combination. Print only the answer.
<Example> <i>Constraints ($\mathcal{C}$):</i> - $[N]=6$-character English word. - Letters $[C_{1..8}] = \text{'e', 'l', 'o', 'd', 'p', 'h', 'i'}$. <i>Possible Answer:</i> hoodie

Table 18: Anagram Scribble.

<Prompt Template ($\mathcal{P}$)>		
Please write a text string without any space by following a set of given rules. Please write only the answer and follow the following criteria:		
- the text has $[C_1]$		
...		
- the text has $[C_\alpha]$		
<Example>		
<i>Constraints ($\mathcal{C}$):</i>		
- $[C_1]$ = 6 English characters		
- $[C_2]$ = 0 uppercase character		
<i>Possible Answer:</i>		
hoodie		
<Possible Rules ($\mathcal{C}_\chi$)>	<Type>	<Repeatable>
- only $[N]$ characters	counting	no
- $[N]$ uppercase characters	counting	no
- $[N]$ lowercase characters	counting	no
- $[N]$ latin character	counting	no
- $[N]$ number digits	counting	no
- $[N]$ number of roman digits	counting	no
- $[N]$ special characters, including '!', '@', '#', '\$', '%', '^', '&', '*'	counting	no
- $[N]$ $[Ch]$ character	counting	yes
- $[S]$ string	string-matching	yes
- the capital city of $[S]$	knowledge	yes
- the continent of $[S]$	knowledge	yes
- a number that equals to $[E_{\text{math}}]$	math	yes
- a number that equals to $[E_{\text{word}}]$	math	yes
<Parameters>		
· $[N] \in \mathbb{Z}^+$; $[Ch] \in \{ 'A' \dots 'Z', 'a' \dots 'z' \}$;		
· $[S]$ is a random English word;		
· $[E_{\text{math}}]$ is an arithmetical expression written in number and symbols, e.g. "4 + 2";		
· $[E_{\text{word}}]$ is an arithmetical expression written in words, e.g. "four plus two";		

Table 19: Password Game.

<Prompt Template (P)> You are given a text [S] Your job is to put some valid parenthesis brackets in the text such that: - [W₁] is inside a [B₁] bracket ... - [W_N] is inside a [B_N] bracket The open and close parenthesis for block is [], curly is { }, round is (), and angle is < >. The bracket depth must be [D] and print only the answer <Example> <i>Constraints (C):</i> The text is [S] = 'fabuloustextgames', and [W] = ['games', 'text', 'fabulous'] are inside [B] = [round, angle, block] bracket, respectively. Depth must be [D] = 2. <i>Possible Answer:</i> {[fabulous]<text>(games)}
--

Table 20: Bracket Game

<Prompt Template (P)> You are given the following string: [S] Find a substring of exactly [N] characters long that: - Contains [X_{1...α}] - Does not contain [Y_{1...β}] - [Z₁] ... - [Z_γ] Print only the answer. <Example> <i>Constraints (C):</i> - [S] = "hengooserabbitant" - [X_{1...1}] = {'g'} - [Y_{1...2}] = {'i', 'a'} - No complex rules [Z] = ∅ <i>Possible Answer:</i> goo	
<Possible Complex Rules [Z_χ]> - forms a palindrome - has 2 consecutive consonants - does not have 2 consecutive consonants - has 2 consecutive vowels - does not have 2 consecutive vowels - has more vowels than consonants - has less vowels than consonants - has the same amount of vowels and consonants	Mutually Exclusive Group - α α β β γ γ γ

Table 21: String Search.

<Prompt Template ($\mathcal{P}$)>

Given a board size of $[N] \times [N]$, arrange a possible crossword puzzle answer from a list of words. Item in the list can only be used once.

List of words:

- $[W_1]$
- $[W_2]$
- ...

Print only the answer.

<Example>

Constraints ($\mathcal{C}$):

- $[N] = 3$ (3x3 grid)
- $[W_{1..8}] = \{\text{app, all, and, lee, let, pat, pee, pet}\}$

Possible Answer:

app
lee
let

Table 22: Crossword Arranger.

<Prompt Template ($\mathcal{P}$)>

Please solve the $[N] \times [N]$ sudoku with $[V]$ as the values and fill _ with the possible value and only print the answer. Follow the sudoku rule.

$[S_{1,1}] \dots [S_{1,N}] \dots [S_{N,1}] \dots [S_{N,N}]$

<Example>

Constraints ($\mathcal{C}$):

- $[N] = 4$ (4x4 grid)
- $[V] = \{A, B, C, D\}$
- $[S_{1,1}] \dots [S_{N,N}] = \text{"A_CD CD_B _AD_ DCBA"}$

Possible Answer:

ABCD
CDAB
BADC
DCBA

Table 23: Text Sudoku.

<Prompt Template ($\mathcal{P}$)>

You are asked to construct a 2D $[N] \times [N]$ grid, consisting of water tiles (denoted by '.'), land tiles (denoted by '#'), and coconut tree tiles (denoted by 'o'). Coconut tree tiles are also considered as land tiles.

A group of connected land tiles in 4 cardinal directions forms an island.

Your 2D grid must follow the following rules:

- There must be exactly $[K]$ islands.
- The size of each island must be from $[Y_{\min}]$ to $[Y_{\max}]$ tiles each.
- There must be exactly $[L]$ islands that have coconut trees on them.
- There must be exactly $[C]$ total coconut trees.

Print only the answer.

<Example>

Constraints ($\mathcal{C}$):

- $[N] = 6$ (6x6 grid),
- $[K] = 3$ islands,
- island size from $[Y_{\min}]=5$ to $[Y_{\max}]=10$ tiles,
- $[L] = 2$ islands have coconut trees,
- $[C] = 4$ coconut trees in total.

Possible Answer:

```
##...  
#o#...  
.o#...  
....##  
#o#...  
#o#...
```

Table 24: Islands.

<Prompt Template ($\mathcal{P}$)>

Given a set of rules to calculate point, sort the set of words in decreasing order.
When there 2 or more words with same point, sort lexicographically.

Rules:

- $[C_1]$ gets $[P_1]$ points
- add $[P_2]$ points if $[C_2]$
- ...

Words:

- $[W_1]$
- $[W_2]$
- ...

Print only the answer.

<Example>

Constraints ($\mathcal{C}$):

Rules:

- add $[P_1]=1$ point if $[C_1] = \text{there exists 'g' in the word}$
- $[C_1] = \text{word less than 5 characters}$ gets $[P_1]=10$ points

Words:

- $[W_1] = \text{hen}$
- $[W_2] = \text{goose}$
- $[W_3] = \text{rabbit}$
- $[W_4] = \text{ant}$

Possible Answer:

ant
hen
goose
rabbit

<Possible Rules Condition $[Z_x]$ >

- every (vowel | consonant)
- every vowel right after a consonant
- every consonant right after a vowel
- every pair of consecutive vowels
- every $[N]$ consecutive vowels
- every pair of consecutive consonants
- every $[N]$ consecutive consonants
- word that has exactly $[N]$ characters
- word not equal to $[N]$ characters
- word (more | less) than $[N]$ characters
- word more than $[N_1]$ characters and less than $[N_2]$ characters
- word (more | less) than $[N_1]$ characters but not equal to $[N_2]$ characters
- word (starts | ends) with ' $[S]$ '
- word starts with ' $[S_1]$ ' and ends with ' $[S_1]$ '
- there exists ' $[S]$ ' in the word
- there exists exactly $[N]$ ' $[S]$ ' in the word

<Type>

Counting
Pattern
Pattern
Pattern
Pattern
Pattern
Pattern
Counting
Counting
Counting
Counting
String-search
String-search
String-search
String-search

<Parameters>

- $[N] \in \mathbb{Z}^+$;
 - $[S]$ is any character sequence;
-

Table 25: Ordering Text

<think>
Okay, so I need to create a 3x3 crossword puzzle using the given list of words: age, ago, dub, man, map, nor, per, pop. Each word *can only be used once*. Let me think about how to approach this.

First, I should list out the words and their lengths. All of them are three letters, which fits perfectly into a 3x3 grid. That's good because *each word will occupy* a row or a column.

I should consider how these words can intersect. Since it's a crossword, the words will share letters where they cross. Let me look for words that share common letters, especially in the middle, because that's where they'll intersect in a 3x3 grid.

Looking at the words:

- age
- ago
- dub
- man
- map
- nor
- per
- pop

...

Wait, let's try to place "nor" somewhere. "Nor" is N, O, R. Maybe place it vertically in column 3: N, O, R.

So column 3: N, O, R.

Then, row 1: _, _, N
Row 2: _, _, O
Row 3: _, _, R

Now, let's see. Maybe "age" is vertical in column 2: A, G, E.

So column 2: A, G, E.

Then, row 1: _, A, N
Row 2: _, G, O
Row 3: _, E, R

Now, let's see what's left: ago, dub, map, man, per, pop.

Looking at row 1: _, A, N. Maybe "man" can go here: M, A, N. So row 1: M, A, N.

Then, column 1: M, _, _. Maybe "map" vertically: M, A, P. But row 1, column 1 is M, so row 2, column 1 would be A, and row 3, column 1 would be P. So row 2: A, G, O; row 3: P, E, R.

Now, let's check the words used: "nor" in column 3, "age" in column 2, "man" in row 1, "map" in column 1. Remaining words: ago, dub, per, pop.

Looking at row 2: A, G, O. That's "ago", which is in our list. So row 2: A, G, O.

Now, row 3: P, E, R. That's "per", which is in our list.

So the grid would look like:

Row 1: M, A, N
Row 2: A, G, O
Row 3: P, E, R

But wait, "map" is M, A, P, but in this configuration, column 1 is M, A, P, which is "map". So that works.

Now, let's check all words used: man, ago, map, per, nor. Wait, "nor" is N, O, R, but in this grid, column 3 is N, O, R, which is "nor". So that's correct.

But we still have "dub" and "pop" left. Hmm, that's a problem because we've used all the words except those two, but they don't fit into the grid.

Wait, no, let's count: we have 8 words, and the grid uses 5 words: man, ago, map, per, nor. So we're missing "dub" and "pop". That means this configuration *doesn't use all* the words, *which is required*.

...

Table 26: DeepSeek-R1-Distill 14B Hallucinated despite getting the correct answer along the way and ended up changing the answer to the wrong one.

Game	Feedback
Anagram	Your answer must be exactly <int> characters long
Scribble	Your answer must only contain the characters provided
	Your answer must not contain repeated characters
	Your answer is not a valid English word
Password	<str _{prediction} > is not satisfying this rule: <str _{rule} >.
Bracket Game	You are not allowed to change the character sequence of base text <string>
	There is a closing bracket without an open bracket
	The depth of the bracket is <int>. The expected depth is <integer>
	The text '<str _{rule_text} >' is not found in your answer.
	The text '<str _{rule_text} >' is not inside any <'block' 'curly' 'round' 'angle'>
	bracket <char _{bracket_open} > <char _{bracket_close} >
String Search	<str _{answer} > is not <int> characters long.
	<str _{answer} > does not exist in <str _{input_text} >.
	<str _{answer} > does not have 2 consecutive consonants
	<str _{answer} > has 2 consecutive consonants
	<str _{answer} > does not have 2 consecutive vowels
	<str _{answer} > has 2 consecutive vowels
	<str _{answer} > has less or equal vowels than consonants
	<str _{answer} > has more or equal vowels than consonants
	<str _{answer} > does not have the same amount of vowels and consonants
	<char> does not appear in <str _{answer} >.
	<char> exists in <str _{answer} >.
	<str _{answer} > is not a palindrome.
Crossword Arranger	Mismatch answer length found!! Expected size of <integer>, got <integer>.
	Mismatch answer word found!! <'Horizontal' 'Vertical'> word <string> is not in the word set.
Text Sudoku	There are unfilled cells
	Your answer is wrong in shape, it should be <int>x<int> sudoku.
	There are unrecognized characters, or possibly unfilled cells.
	One or more characters are replaced
	Invalid
Islands	2D grid is not <int> x <int>. (<int _{pred} > x <int _{pred} >)
	2D contains invalid character (<char>)
	There must be exactly <int> islands, but you provided <int> islands
	The size of each island must be from <int> to <int> tiles
	There must be exactly <int> islands that have coconut trees on them
	There must be exactly <int> total coconut trees.
Ordering Text	Your answer is too short. There should be <int> items.
	<str _{answer} > is not supposed to be the <str _{ordinal_number} > word in the order.

Table 27: List of Feedback.